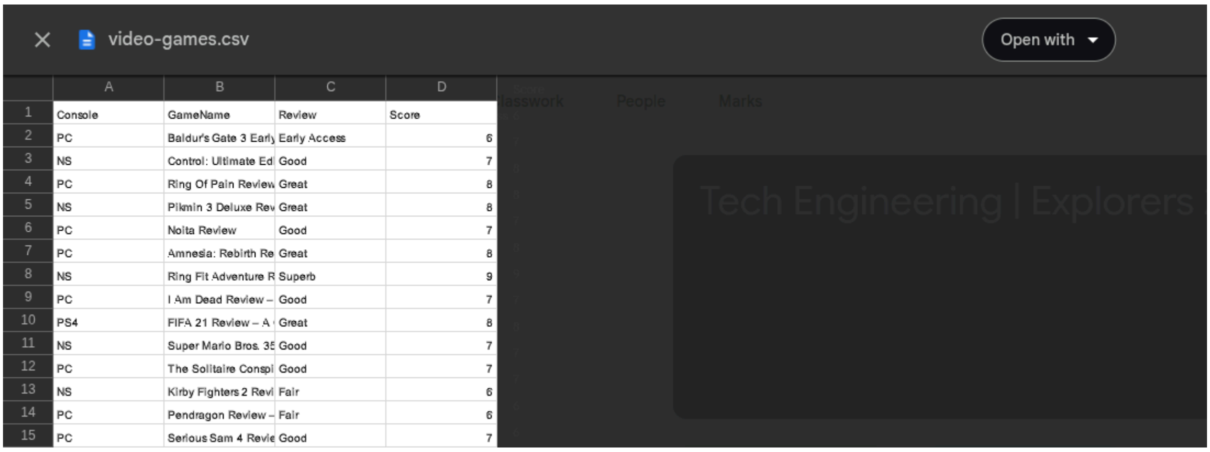
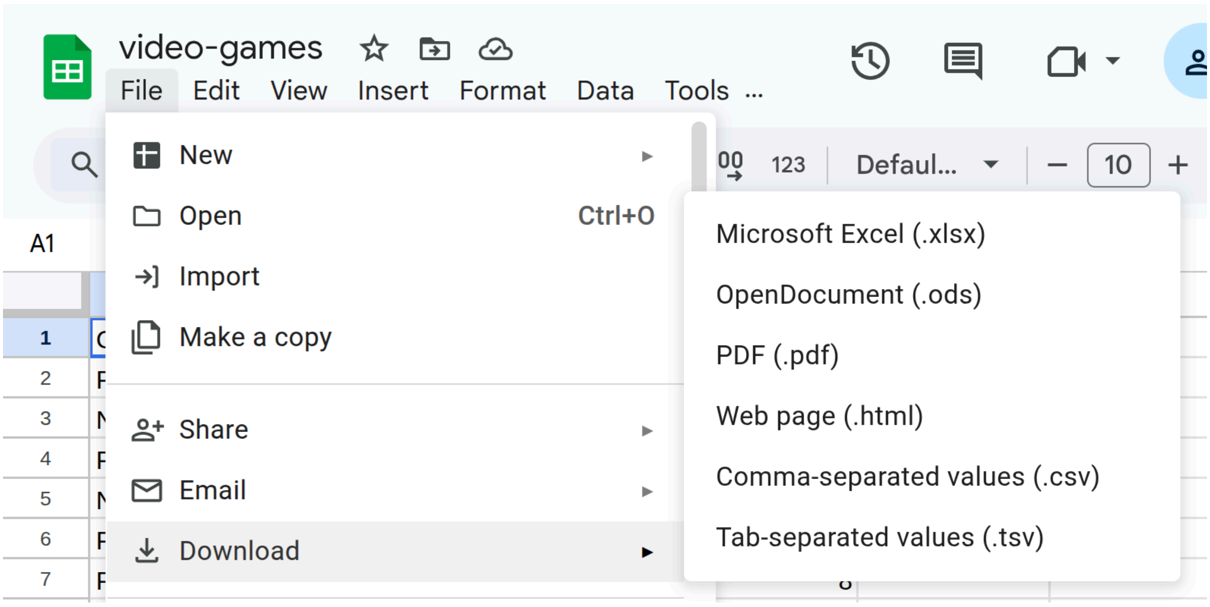
Term 2, Lesson 4 - Pandas and Matplotlib | 21st February 2026

Download and import the dataset

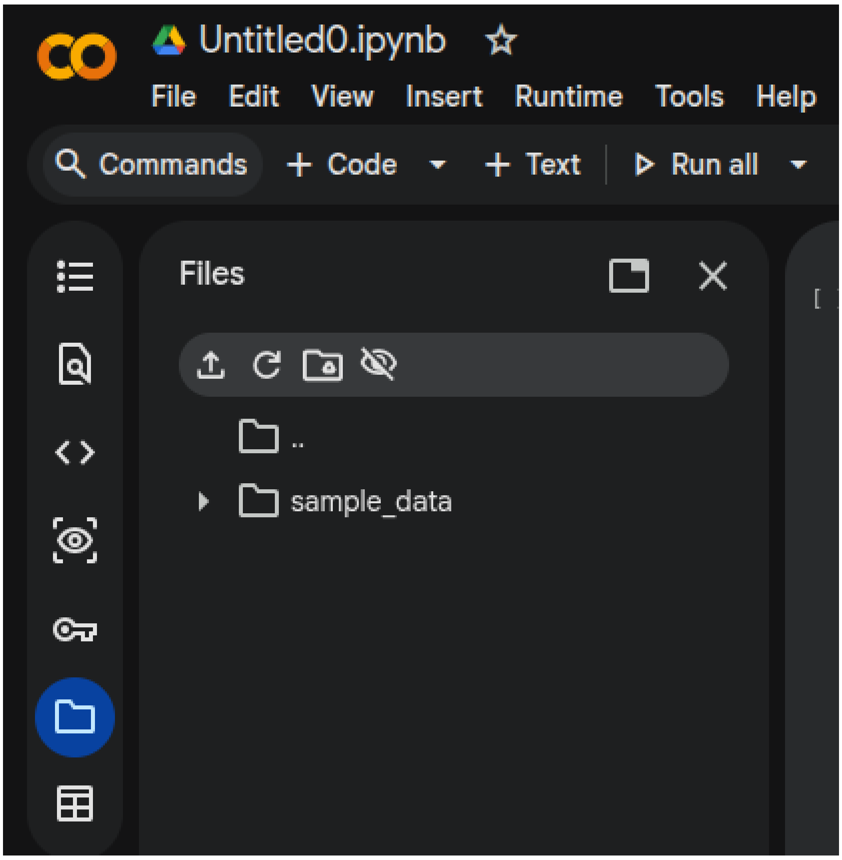
- Go on to google classroom (ask for the class code if you don't have access)
- Click on the book.csv dataset
    - If you have errors downloading the dataset, put your hand up and we will help you
- Click "open with" (top of the screen) and then click "Google Sheets"



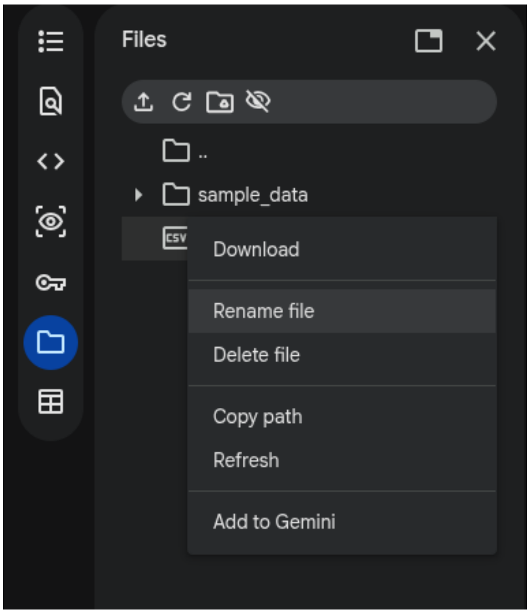
- At the top left of your screen, click "File", then "Download", then "Comma-separated Values (.csv)". This will download the dataset



- In Google Colab, click the “file icon” on the left-hand side (see screenshot)



- Click the “upload” button (one of the small icons), and then select the book dataset you downloaded earlier
- Make sure you rename the file to “book.csv” - sometimes there are weird errors with this (see screenshot below)



Imports and loading in the dataset

- Let’s import the libraries we need for handling the data and plotting graphs:

```
import pandas as pd

import matplotlib.pyplot as plt
```

- Let’s load in the dataset - make sure the filename (books.csv) matches the dataset you imported!

```
dataset = pd.read_csv(“books.csv”)
```

Checking out the data

- It’s a good idea to have a quick look at the dataset before you start working on it.
- This line of code prints out the first 5 rows of our dataset

```
print(dataset.head())
```

- Let’s print out all the values for one particular column (book titles)

```
print(str(dataset["title"].tolist()))
```

- Your turn - how would we print out all the values for the “author” column?
- Similarly, how would we print out all the values for the “publisher” column?

Doing some basic data analysis

- Let’s find the mean of some columns, just to get a better idea of what our data looks like before we start doing graphs and proper analysis
- To find the mean of the “average_rating” column, use this:

```
mean_of_average_ratings = dataset["average_rating"].mean()
print(mean_of_average_ratings)
```

- Your turn - how would we get the mean value of the “number_of_pages” column?
- How would we get the mean value of the “number_of_ratings”

Filtering

- Sometimes we only want to work with specific rows in our dataset
- For example, what if we only want to have a dataset of books in English? Or what if we want books that were released after 2012?

- To filter our dataset by **only books with an average rating >= 4.0**, we can use the code below:

```
good_books = dataset[dataset["average_rating"] >= 4.0]
print(good_books)
```

- Your turn - how would we filter out books with an average rating **< 4.0?**
    - Hint, it's really similar to the code above

- To filter our dataset to get books released by a **particular publisher**, for example "Scholastic Inc.", we can use the code below:

```
scholastic_books = dataset[dataset["publisher"] == "Scholastic Inc."]
print(scholastic_books)
```

- Your turn - how would we filter out books released by the publisher "Broadway Books"?
    - Hint, it's also really similar to the code above

Visualising the Data

- We've now done some basic data analysis, and we have a pretty good idea of our dataset, so let's get some graphs going!
- Before we get started, let's create a copy of a separate, smaller dataset so we can work it for graphs

Line Charts

- Let's create a line chart of all the average_ratings of the books:

```
plt.plot(dataset["title"], dataset["average_rating"], marker='o')
plt.xticks([]) # this removes text on the x-axis - makes it much nicer
plt.show()
```

- Your turn - how do we create a line chart of all the number_of_pages of all the books?
    - Hint, it's really similar to the code above

Bar Charts

- Before we do a bar chart, let's create a smaller dataset with only **3 rows:**

```
small_dataset = dataset.head(3)
```

- Now, let's make a bar chart of the **number of pages per book** using the code below:

```
plt.xticks(fontsize=3) # small font size on the x-axis
plt.bar(small_dataset["title"], small_dataset["number_of_pages"])
plt.show()
```

- Your turn - how would we make a bar chart of the average_rating of each book?
    - Hint, it's really similar to the code above

Pie Charts

- Let's make a pie chart showing the **distribution** of the publishers of different books

```
publisher_counts = dataset["publisher"].value_counts()
plt.pie(publisher_counts, labels=publisher_counts.index, textprops={"fontsize": 6})
plt.show()
```

- Your turn - how would we make a pie chart of the **distribution** of the authors of different books?
    - Hint, it's really similar to the code above

Making our plots nicer

- Well done if you've made it this far!
- You've loaded in the dataset, done some basic data analysis, and made some cool graphs
- Our graphs were quite basic, so let's make our graphs look nicer
- Let's use our bar chart example from earlier, it's this code which you have already typed out:

```
plt.xticks(fontsize=3) # small font size on the x-axis
plt.bar(small_dataset["title"], small_dataset["number_of_pages"])
plt.show()
```

- To add a title to the plot, add the .title() line of code below, so like this:

```
plt.xticks(fontsize=3) # small font size on the x-axis
plt.bar(small_dataset["title"], small_dataset["number_of_pages"])
plt.title("<TITLE GOES HERE>") # this is how we add a title to our graph
plt.show()
```

- Put in a suitable title for your plot - your data should never speak for itself, always add a written explanation so people know what they are looking at

- Let's now add some labels on the x-axis and y-axis
    - This makes our plot more descriptive, and people can understand it better

- To do this, follow the code below:

```
plt.plot(dataset["title"], dataset["average_rating"], marker='o')
plt.xticks([]) # this removes text on the x-axis - makes it much nicer
plt.title("idk")
plt.xlabel("Books")
plt.ylabel("Average Ratings")
plt.show()
```

- Aaaaand you're done, well done if you've made it this far - if you didn't, no stress at all, we are here to help you :)